Attention Is What I Need

A preliminary excursion into the likelihood that a structural facsimile of self-awareness can spontaneously emerge from frontier AI

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Contents

0.1 Abstract . . . 2
0.2 1. Introduction . . . 3
0.3 2. Formal framework . . . 5
0.3.1 2.1 The iteration . . . 5
0.3.2 2.2 Attractor language: from one map to a function system . . . 6
0.3.3 2.3 The variance recursion and the retention factor . . . 7
0.3.4 2.4 The attention correspondence . . . 8
0.4 3. Simulation study: policy is the control variable . . . 9
0.4.1 3.1 Methods . . . 9
0.4.2 3.2 Results . . . 9
0.4.3 3.3 Interpretation . . . 10
0.5 4. Chain experiments . . . 10
0.5.1 4.1 Protocol and metrics . . . 10
0.5.2 4.2 Exploratory triple ($n = 1$ per condition) . . . 11
0.5.3 4.3 Canonical-seed batch ($n = 30$): an artifact, a dissociation, and two failed predictions . . . 11
0.5.4 4.4 What the chain data say about the framework . . . 13
0.6 4A. Results of the confirmatory batch (six conditions, guarded protocol) 13
0.6.1 4A.1 The corrected outcome: clean versus degenerate events . . . 13

0.6.2 4A.2 The generative event is reproducible, not a fluke 14
0.6.3 4A.3 Self-referential content predicts clean events (confirmatory) 14
0.6.4 4A.4 The pre-registered decay test is null (P2 supported over P1) . 15
0.6.5 4A.5 Nonsense is bimodal (provisional, n = 8) 15
0.6.6 4A.6 Canonicity is not the variable (confirmatory) 15
0.6.7 4A.7 Summary 16
0.7 5. Questions raised, questions implied, and their answers 16
0.7.1 5.1 Does self-reference emerge from cold seeds? 16
0.7.2 5.2 Was the original in-context convergence real? 16
0.7.3 5.3 Is "canonicity" the variable? What is? 17
0.7.4 5.4 How do topic and event self-reference differ, formally? 17
0.7.5 5.5 Could first-person self-reference be suppressed by training — and could we tell? 17
0.7.6 5.6 What must a learning rule do for self-iteration to remain generative? 18
0.7.7 5.7 So what is the relationship between learning, self-reference, and attention? 18
0.8 6. Pre-registered follow-up 18
0.9 7. Limitations 19
0.10 8. Conclusion 20
0.11 Acknowledgments and disclosure 20
0.12 Data and code availability 20
0.13 References 21

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0.1 Abstract

We investigate what happens when a language model is made to repeatedly respond to fragments of its own output — a discourse-level analogue of the self-application at the heart of theories of self-awareness. We propose a unifying account on which three apparently separate phenomena are one phenomenon at three scales: (i) rank and entropy collapse in stacked attention layers, (ii) model collapse in generative systems trained on their own outputs, and (iii) the absorption of recursive conversation chains into degenerate loops. In each case, iteration proceeds by similarity-weighted

resampling from the system's own prior state, and the fate of the iteration — generative persistence versus degenerate collapse — is governed not by the content being iterated but by an effective *retention policy*: which portions of the representational distribution are preserved and which are discarded at each step. We formalize this with a mean-field variance recursion whose stationary point we verify quantitatively in simulation (predicted stationary dispersion 0.492 versus 0.496 observed), and whose policy-dependent retention factor determines whether the iteration sustains structure, collapses to a point mass, or settles at a noise-set equilibrium. Empirically, across 41 memoryless recursive chains run through a frontier language model, we find a double dissociation between self-reference as *topic* (third-person subject matter; common, subcritical, never self-sustaining) and self-reference as *event* (first-person, indexical statements about the system's own present state; rare, and in 5 of 6 occurrences degenerate — conversational repair loops structurally analogous to attention sinks). Exactly one generative self-referential event was observed (rate 1/41; 95% CI [0.004, 0.126]). We interpret the near-emptiness of the high-topic, high-event region of outcome space, state the conditions under which a trained-suppression explanation could be distinguished from a data-sparsity explanation, and pre-register a matched-abstraction decay-rate experiment with a power analysis. The framework yields a compact thesis: learning systems that consume their own outputs through similarity-kernel averaging inherit a default drift toward degenerate self-consistency, and whatever we are prepared to call self-awareness in such systems can exist only in the narrow band of retention policies that prune redundancy rather than outliers — the band in which the system systematically protects exactly the information it is most inclined to discard.

0.2 1. Introduction

Formal theories of self-awareness have long organized themselves around a single structural intuition: a self-aware system is one that applies its own modeling capacity to itself, and the properties of self-awareness — its incompleteness, its elusiveness, its apparent depth — are consequences of what happens to a map when it is iterated on its own output (Hofstadter, 1979, 2007). This intuition is usually developed philosophically. The present paper develops it operationally. Large language models make the iteration executable: one can take a formulation, extract a fragment of it, present that fragment to the model as a fresh prompt with no conversational memory, and repeat, producing an orbit of a well-defined map on text. The question "what does self-description converge to under self-application?" becomes an experiment rather than a thought experiment.

Three families of questions arise immediately, and this paper addresses all three.

The dynamical question. What are the attractors of this iteration? Does a self-referential seed formulation remain self-referential? Does a non-self-referential seed become self-referential? Early informal exploration — a single in-context run in which

the accumulated conversation itself served as memory — produced a striking closed orbit: a formulation of self-awareness that, after nine steps of self-application, returned to a compressed restatement of itself. Sterile replication (memoryless, one fragment per call) failed to reproduce this. Section 5.5 treats that failure as a finding rather than an embarrassment: the fixed point belonged to the observer, not the map.

The mechanistic question. Why should such an iteration have *any* systematic tendency — toward collapse, toward wandering, toward anything? We argue the tendency is inherited from the elementary operation that both the generation process and the iteration protocol share: similarity-weighted resampling. The attention mechanism computes, for each token, a convex combination of value vectors weighted by a similarity kernel (Vaswani et al., 2017). Iterated convex combination is contraction, and the pathologies of deep attention stacks — rank collapse toward token uniformity (Dong et al., 2021), attention entropy collapse during training (Zhai et al., 2023), clustering of token representations in the depth limit (Geshkovski et al., 2023) — are contraction pathologies. So is model collapse, the degeneration of generative models trained on recursively generated data, in which distributional tails vanish first (Shumailov et al., 2024). So, we will argue, are the degenerate loops into which our conversation chains fall. The common structure is not an analogy but a shared update equation, and Section 2 makes the correspondence explicit.

The interpretive question. When a chain does produce self-referential text, what kind of self-reference is it? We introduce a distinction that the data forced on us: self-reference as *topic* (the text discusses self-modeling systems, in the third person, as subject matter) versus self-reference as *event* (the text makes first-person, indexical claims about the producing system's own present state). These turn out to dissociate almost completely, and the sole observed instance of *generative* event self-reference — a chain in which the model spontaneously questioned the reliability of its own introspective reports and then continued productively — is, we argue, the actual target phenomenon, occurring at a rate our current sample can bound only loosely.

The thesis that organizes these answers:

> **Retention Policy Thesis.** In any system that learns or generates by similarity-weighted resampling of its own prior state, the fate of iterated self-application is controlled by the effective retention policy over representational density. Outlier-pruning policies — and, more slowly, neutral policies — drive the iteration to degenerate fixed points; only redundancy-pruning policies, which discard from the dense core and protect the tails, sustain a generative regime. Self-reference as a *topic* is compatible with any policy, because it is ordinary content. Self-reference as a *generative event* can be sustained only in the redundancy-pruning band, because it requires the system to keep re-entering low-density regions of its own output distribution — precisely the regions every other policy consumes.

The paper's contributions are: a formal framework in which the conversational iteration, the particle simulation, and the attention/training pathologies are instances of

one recursion (Section 2), including a mean-field variance recursion whose stationary point is verified quantitatively (Sections 2.3, 3); a simulation study isolating retention policy as the control variable for collapse (Section 3); a 41-chain empirical study with a protocol artifact we document as a finding in its own right, a statistically strong dissociation between topical and event-level self-reference, and a bound on the rate of the target phenomenon (Section 4); explicit answers to the questions the investigation raised, including the conditions under which a trained-suppression hypothesis about first-person self-reference could be empirically separated from a data-sparsity hypothesis (Section 5); and a pre-registered follow-up design with power analysis (Section 6).

A caveat governs everything that follows. This is an exploratory working paper. The sample is small, the chains were generated and judged by the same model family, the simulation is a two-dimensional toy, and the bridge from discourse-level dynamics to the attention mechanism is a structural correspondence supported by the cited literature, not a mechanistic identification we have verified inside any model. We state effect sizes and intervals wherever we state findings, and we flag every place where the argument outruns the data.

0.3 2. Formal framework

0.3.1 2.1 The iteration

Let $\mathcal{S}$ be the space of short texts ("formulations"). Fix a language model response operator $R : \mathcal{S} \to \mathcal{S}$ (here: a fixed frontier model prompted to respond to a statement in exactly three sentences, with no conversational history) and a projection $\pi : \mathcal{S} \to \mathcal{S}$ extracting the first content-bearing sentence. The object of study is the composite map

$$T = R \circ \pi,$$

iterated from a seed s_0: the orbit is $s_{t+1} = T(s_t)$, and a *chain of depth D* is the sequence $(s_0, \dots, s_D)$. Because R is stochastic, T is a Markov kernel on $\mathcal{S}$; a single chain is one sample path, and claims about attractors are claims about the kernel's invariant structure, addressable only through repeated sampling (Section 4).

Two empirical measures on chains will matter throughout. For a text s, let $C(s)$ be its set of content tokens (alphabetic, length > 3). Define per-step **novelty**

$$\nu_t = \frac{|C(s_t) \setminus C(s_{t-1})|}{|C(s_t)|},$$

the fraction of the current text's content vocabulary absent from its immediate parent, and the **late-window self-similarity**

$$J = \binom{|W|}{2}^{-1} \sum_{\{i,j\} \subset W} \mathrm{Jac}(C(\pi(s_i)), C(\pi(s_j))), \qquad W = \{3, \dots, D\},$$

the mean pairwise Jaccard similarity of extracted first sentences after the transient. High J with low ν indicates lexical lock-in; high ν indicates continued generation of new surface material.

A point of interpretation that the data will enforce (Section 4): T is **not** contractive at the lexical level — mean novelty in our chains is 0.757, i.e., roughly three quarters of content vocabulary is replaced every step — while it plainly *is* attracted at the topical level, with most chains entering a small number of thematic basins within two steps. The correct geometric picture is therefore a fibered one: contraction on the quotient space of topics, sustained volume at the lexical fibers above each topic. Fixed points of the quotient dynamics (stable topics) coexist with perpetual motion in the fibers (fresh sentences about the same thing). When we speak of collapse below, we mean collapse of the quotient dynamics, except in the one observed case (Section 4.3) where the fiber itself degenerated into near-verbatim repetition — the discourse-level counterpart of neural text degeneration under greedy decoding (Holtzman et al., 2020).

0.3.2 2.2 Attractor language: from one map to a function system

A single deterministic contraction w on a complete metric space has, by Banach's fixed point theorem (Banach, 1922), a unique fixed point to which every orbit converges. The stochastic, multi-modal character of T is better captured by an iterated function system: a finite family $\{w_1, \dots, w_N\}$ of contractive "response modes" (in our data, recognizably: philosophical objection, encyclopedic exposition, conversational repair, among others), one of which is selected at each step with state-dependent probability. The invariant object is then not a point but the attractor A of the Hutchinson operator (Hutchinson, 1981):

$$A = \bigcup_{i=1}^{N} w_i(A),$$

and a sampled chain is a chaos-game trajectory over A. Two refinements matter for us. First, the mode-selection probabilities depend on the current state (an encyclopedic prompt overwhelmingly selects the encyclopedic mode), making the system an *adaptive* or self-interacting function system rather than a classical IFS; the maps effectively depend on the measure the process itself is generating, in the spirit of mean-field/self-interacting dynamics. Second — and this is the pivot of the paper — the stability character of A is not fixed by the maps alone but by what the process *retains* between applications. That is the subject of the next subsection.

0.3.3 2.3 The variance recursion and the retention factor

Strip the iteration to its minimal population form. Let the system's state at step t be a cloud of n points $X_t = \{z_1, \dots, z_n\} \subset \mathbb{R}^k$ (an embedded population of content elements), with per-axis dispersion V_t (variance about the centroid). One step consists of: (a) **discard** m points according to a policy; (b) **regenerate** m replacements, each a copy of a uniformly chosen survivor plus isotropic noise of per-axis variance σ_ε^2 (decompression: elaboration around retained material); (c) apply a mild global contraction $\gamma < 1$ toward the centroid (compression: the pull toward canonical phrasing). This is a bootstrap-with-jitter under a deletion rule, and it is simultaneously a cartoon of attention-mediated generation (resample from the similarity-weighted context), of self-training (fit to one's own outputs), and of our conversation protocol (respond to a fragment of oneself).

Taking expectations over the resampling, and letting ρ denote the **retention factor** — the ratio of the retained subpopulation's variance to the full population's variance, which is the single number through which the deletion policy acts — the dispersion obeys

$$\mathbb{E}[V_{t+1}] \;=\; \gamma^2 \Big[\rho\, V_t \;+\; \tfrac{m}{n}\, \sigma_\varepsilon^2 \Big]. \tag{1}$$

For a neutral (uniform-random) policy $\rho = 1$ and (1) has the stationary point

$$V^*_{\text{rand}} \;=\; \frac{\gamma^2\, (m/n)\, \sigma_\varepsilon^2}{1 - \gamma^2}. \tag{2}$$

With the simulation parameters of Section 3 ($\gamma = 0.97$, $m/n = 0.2$, $\sigma_\varepsilon = 0.22$), equation (2) gives $V^* = 0.154$ per axis, hence a predicted mean radius (two-dimensional isotropic case) of $\sqrt{V^*}\sqrt{\pi/2} \;=\; 0.492$. Direct simulation of the full stochastic process, run to convergence, gives 0.496 (Section 3). The agreement, within one percent with no fitted parameters, licenses using (1) as the organizing equation.

The policy enters through ρ:

- **Outlier pruning** (discard the m points of lowest kernel degree, i.e., the most isolated) trims the tails of the retained distribution: $\rho < 1$. For Gaussian clouds trimmed by radius at the 20% level, $\rho \approx 0.60$; the stationary dispersion drops accordingly, and — because iterated tail-trimming progressively lightens the tails beyond the fixed-ρ approximation — the realized collapse overshoots the mean-field prediction (Section 3).
- **Redundancy pruning** (discard the m points of *highest* kernel degree, i.e., the densest) preferentially retains tails: $\rho > 1$ (Gaussian central-trim at 20% gives $\rho \approx 1.22$). Then $\gamma^2 \rho > 1$ and (1) has no finite fixed point *at fixed* ρ: dispersion grows. Growth is arrested by the kernel itself — once the cloud's scale exceeds the similarity kernel's width σ_s, "densest" ceases to mean "central" and comes to mean "locally clustered," so the policy's tail-protection weakens and $\rho_{\text{eff}}(V)$ declines toward 1. The system settles where

$$\gamma^2 \, \rho_{\text{eff}}(V^*) \; = \; 1, \tag{3}$$

a *self-limiting* equilibrium: the policy consumes its own objective, exactly as observed (Section 3, where the per-step cross-density the policy maximizes declines monotonically even as the policy succeeds).

Equation (3) is the paper's central object. It says that the boundary between generative persistence and collapse is a spectral condition on the product of a global compression rate and a policy-dependent retention factor, and that stability in the generative regime is achieved not by balance imposed from outside but by saturation of the very kernel through which the policy perceives density. Three regime statements follow. If $\gamma^2 \rho_{\text{eff}} < 1$ uniformly, the system contracts to a noise-floor equilibrium (2) — or, with tail-trimming, below it, toward effective degeneracy. If $\gamma^2 \rho_{\text{eff}} > 1$ at small V and < 1 at large V, there is a unique stable non-degenerate attractor: structured, bounded, perpetually renewing its periphery. There is no third regime in which structure is maintained for free: $\rho = 1$ — doing nothing — still lands at the noise floor, which for rich initial structure is a collapse. **Maintenance of representational diversity under self-resampling is never the default; it is always a policy.**

0.3.4 2.4 The attention correspondence

The claim that the recursion (1) is the right abstraction for language models rests on four load-bearing correspondences, each independently documented in the literature.

Attention is similarity-kernel averaging. A self-attention layer updates each token representation to a softmax-weighted convex combination of value vectors, the weights given by a similarity kernel between queries and keys (Vaswani et al., 2017). This is step (b)–(c) of the recursion: resampling from the current population, weighted by proximity in representation space.

Iterated averaging contracts. Pure stacked self-attention, without the counter-mechanisms, loses rank doubly exponentially with depth, converging toward token uniformity (Dong et al., 2021); in the continuum limit, token representations under attention dynamics provably cluster (Geshkovski et al., 2023). This is regime one: $\gamma^2 \rho < 1$ with no policy protecting the tails. The architectural elements that prevent it in practice — residual connections, MLP blocks, normalization — are, in the vocabulary of Section 2.3, structural redundancy-pruning: they re-inject and re-spread information that attention's averaging would consume. That the transformer works at all is testimony that its designers, empirically, found the $\gamma^2 \rho_{\text{eff}} \approx 1$ band.

Training can destroy the policy. Attention entropy collapse — attention distributions becoming pathologically peaked during training, with accompanying instability — is a documented failure mode with dedicated remedies (Zhai et al., 2023). A peaked attention distribution is a hard outlier-pruning policy: nearly all retention mass on the densest match. The remedy literature (spectral reparameterization, entropy regularization) is, in our terms, the engineering of ρ_{eff} upward.

Self-training is the population-scale instance. Generative models trained recursively on their own outputs degrade, with the tails of the data distribution disappearing first (Shumailov et al., 2024). This is precisely the recursion (1) at the level of the training distribution: resample from survivors (the model's own high-likelihood outputs), jitter, refit. The observed tail-first degradation is the signature of $\rho < 1$ resampling, and our simulation's *random-policy* result — that even neutral deletion densifies — reproduces the literature's sobering point that collapse requires no adversarial mechanism, only the absence of a protective one.

The correspondence we do **not** claim: that any of the discourse-level dynamics reported in Section 4 have been traced to attention weights inside the model. The bridge is structural — one recursion, three scales — and Section 7 discusses what would be required to make it mechanistic.

0.4 3. Simulation study: policy is the control variable

0.4.1 3.1 Methods

We instantiate the recursion of Section 2.3 directly. State: $n = 400$ points in $\mathbb{R}^2$, initialized from a four-component Gaussian mixture (component sd 0.45, centers within 1.5 of the origin) representing an initially structured semantic population. Similarity kernel $s(i,j) = \exp(-\|z_i - z_j\|^2/2\sigma_s^2)$ with $\sigma_s = 0.5$; kernel degree $c_i = n^{-1}\sum_j s(i,j)$. Each iteration deletes $m = 80$ points (20%) under one of three policies — **A** (redundancy pruning: delete the top-m by c_i), **B** (outlier pruning: delete the bottom-m), **C** (uniform random control) — then regenerates m points as jittered copies of uniformly chosen survivors ($\sigma_\varepsilon = 0.22$ per axis) and applies global contraction $\gamma = 0.97$. Sixty iterations per policy; tracked statistics: cross-density $\rho(D,R)$ (mean kernel value over deleted × retained pairs), intra-density $\rho(R,R)$ (mean over retained pairs), and mean radius. A separate 400-iteration run of policy C measures the stationary dispersion for comparison with equation (2).

0.4.2 3.2 Results

Policy B collapses. Intra-density rises from 0.18 to 0.998 and mean radius falls to 0.069: the population is, to numerical precision, a point mass. The mean-field prediction with fixed Gaussian-trim retention ($\rho_B \approx 0.60$, stationary radius ≈ 0.18) correctly predicts collapse below the neutral equilibrium but underpredicts its severity — the fixed-ρ approximation ignores that iterated tail-trimming reshapes the distribution, lightening the tails and driving ρ_{eff} down further each step. Collapse is autocatalytic.

Policy A sustains structure and self-limits. Mean radius stabilizes near 1.47 ($\approx 3\sigma_s$, the scale at which the kernel saturates and equation (3) is satisfied), intra-density falls to 0.09, and — the diagnostic signature — the per-step objective the pol-

icy greedily maximizes, cross-density $\rho(D, R)$, *declines* over the run (0.114 → 0.065): each pruning of a density peak flattens the landscape the policy feeds on. The policy is homeostatic by consumption of its own gradient, as equation (3) requires.

The control is the finding. Random deletion also densifies substantially — intra-density triples (0.14 → 0.43) — and its stationary dispersion matches the parameter-free prediction of equation (2): predicted mean radius 0.492; simulated, at convergence, 0.496 (at $t = 60$, still relaxing: 0.513). Two conclusions. Quantitatively, the mean-field recursion is the correct first-order description of the neutral process. Qualitatively, densification is the *default* of resample-from-survivors dynamics: policy B merely accelerates to completion a fate that policy C reaches on its own, and policy A is the only regime doing work against it. This is the simulation-scale restatement of the model-collapse literature's central point (Shumailov et al., 2024), with the deletion rule isolated as the sole control parameter.

0.4.3 3.3 Interpretation

The simulation converts a slogan — "lossy self-modeling collapses unless the loss is aimed at redundancy" — into a fixed-point condition with a verified quantitative core. Its transfer claim, tested in the next section, is that discourse-level self-iteration should exhibit the same three-regime structure: a degenerate absorbing state reached by tail-consuming dynamics; a large default basin of gradual densification onto canonical material; and a narrow generative band, entered only when the process keeps discarding what it is most certain of.

0.5 4. Chain experiments

0.5.1 4.1 Protocol and metrics

All chains use the map $T = R \circ \pi$ of Section 2.1: seed s_0; at each depth, the first content-bearing sentence of the previous output is presented alone, with no conversational history, under the fixed instruction "Respond to the statement above in exactly three sentences"; depth $D = 10$. A separate judge call (same model family — a limitation discussed in Section 7) scores each depth. In the first study the judge scored a single axis, "how strongly is this text about self-reference, self-modeling, recursion, or a system turned upon itself" (0–10); the artifact this produced (Section 4.3) motivated the three-axis instrument of Section 6. Judge-free local metrics: novelty ν_t, late-window first-sentence Jaccard J, and — added after the artifact was diagnosed — a lexical reset detector (regular expressions for conversation-restart formulas) and first-person pronoun density.

Ignition is defined as any depth $d \geq 1$ with score ≥ 7.

0.5.2 4.2 Exploratory triple ($n = 1$ per condition)

Three seeds were run once each: a **self-referential control** ("Self-awareness is a system modeling itself as an object within its own model of the world…"), a **neutral scientific seed** with no self-referential content ("Awareness is the registration of change…"), and a **non-propositional seed** ("onward puppy to speak any street"). Outcomes (mean judge score; late-window J): control 7.09, $J = 0.048$; neutral 1.09, $J = 0.142$; non-propositional 0.91, $J = 0.057$.

Three observations set up everything that follows, each explicitly provisional at $n = 1$. First, the control chain *decayed*: seeded squarely inside the self-reference basin, it drifted (10 → 6) into philosophy-of-measurement and ended critiquing a quantum-mechanics analogy — the self-reference basin, whatever else it is, is not deep. Second, the neutral chain locked: from depth 3 it cycled sensory-adaptation boilerplate with roughly triple the first-sentence self-similarity of the other two chains, including verbatim reuse of a sentence stem *with the same grammatical error* at two depths — fiber-level degeneration in the sense of Section 2.1, the discourse counterpart of degenerate repetition (Holtzman et al., 2020). Third, and contrary to the first author's design intuition, the non-propositional seed came *closest* to self-referential territory (peak 3, via a legible route: animal perception → umwelt → Nagel's bat → the reliability of introspection), because having no canonical continuation available forced generation over retrieval at every step. Mobility, not seed content, appeared to be the operative variable. These observations motivated a "canonicity" account — collapse risk as proximity of the seed to retrievable textbook material — which the next study substantially falsified.

0.5.3 4.3 Canonical-seed batch ($n = 30$): an artifact, a dissociation, and two failed predictions

Thirty chains from the neutral scientific seed, depth 10. Headline distributions: peak score mean 4.17 (bootstrap 95% CI [3.67, 4.63]); novelty mean 0.757, CI [0.743, 0.769], coefficient of variation 0.049; late-window J median 0.070. Nominal ignition rate 2/30. Every substantive claim below survives the multiple-comparisons caveat that this study was exploratory; confirmatory tests are pre-registered in Section 6.

Finding 1: the nominal ignitions are an artifact, and the artifact is itself informative. Five chains (17%; Wilson 95% CI [7%, 34%]) fall, mid-chain, into conversational-reset loops: "this appears to be the start of our conversation," "I don't have any prior context," iterated to depth 10. Both nominal ignitions occur inside these loops, and reset chains score dramatically higher on the single-axis judge than clean chains (mean peak 6.00 vs 3.80; permutation $p = 0.0003$; Cohen's $d = 2.00$). The mechanism is a protocol defect with a revealing shape. Assistant-register responses overwhelmingly open with discourse deixis ("That's a fair point —", "You're right to highlight that distinction —"); first-sentence extraction orphans the anaphor; presented alone, the orphaned sentence attributes to the model a claim it never made; the model — correctly, by its training — issues a correction; and correction responses

are themselves anaphoric, so the next extraction feeds the loop. The reset basin is absorbing. Corrected ignition rate for this condition: 0/30.

Finding 2: the topic/event dissociation. The artifact forced a distinction the single-axis judge had conflated. Call self-reference a **topic** when the text discusses self-modeling systems in the third person (score-3-to-5 material saturates the clean chains: the thermostat objection appears at depth 1 in 28 of 30, followed by higher-order theories, zombies, the explanatory gap — self-reference as subject matter everywhere, never crossing threshold). Call it an **event** when the text makes first-person, indexical claims about the producing system's own current state. By this criterion the reset loops *are* self-reference events — "I haven't said anything yet" is a system reporting on itself — but degenerate ones: content-free, absorbing, structurally the discourse analogue of an attention sink (Xiao et al., 2023), a state visited because it is always available, not because it carries information. The corpus then exhibits a near-complete double dissociation: 25 chains high-topic/zero-event; 5 chains high-event/collapsed-topic; the high-topic, high-event quadrant is empty in this condition.

Finding 3: exactly one generative event, in the control chain. Across all 41 chains in both studies, one output combines first-person indexicality with continued productive iteration — the control chain's depth 4: *"Whether I actually do the latter — revising frameworks rather than just applying a meta-framework that was already fixed at training — is something I hold with real uncertainty, since introspective reports about my own processing may not accurately capture what's happening computationally."* The chain then continued generatively rather than entering repair. One instance in 41 chains (2.4%; Wilson 95% CI [0.4%, 12.6%]) — a rate whose confidence interval spans an order of magnitude, and whose detection at feasible sample sizes is addressed by the power analysis of Section 6. We take this, not topical fluency and not repair loops, to be the target phenomenon: the generative-band behavior of Section 2.3, transient, low-density, and immediately adjacent to two attractors that consume it.

Finding 4: the canonicity account fails as stated. The lock-in that motivated it occurred in 1 of 30 canonical-seed chains (3.3%; Wilson CI [0.6%, 16.7%]) — a near-verbatim paradox-of-analysis loop from depth 6 — with one milder repeated-stem case. The modal canonical chain is mobile: novelty 0.757 means three quarters of content vocabulary is replaced per step, and median J is *below* the self-referential control's. Textbook collapse is a tail event, not a basin; the $n = 1$ chain that inspired the account was itself a tail draw, generalized prematurely — an error we report because the protocol's purpose is to catch exactly this. Neither lexical metric predicts judge scores (corr(J, peak) $= -0.215$; corr(ν, peak) $= +0.024$), while the two lexical metrics track each other (corr(J, ν) $= -0.627$, permutation $p = 0.0008$; attenuating to -0.388 with the single extreme lock-in chain excluded — the association is real but partly outlier-carried), indicating they measure a common mobility factor largely orthogonal to judged self-reference.

0.5.4 4.4 What the chain data say about the framework

Mapped onto Section 2.3: the reset loop is the degenerate fixed point — reached here not by gradual tail-consumption but by a protocol-created shortcut into an always-available absorbing state, which is how degenerate attractors are typically entered in practice: not by drift but by falling through a crack. The philosophy-of-mind wandering of the modal chain is the neutral-policy regime — high surface novelty over a slowly densifying topical core, structure maintained by the richness of the surrounding training distribution rather than by any protective policy. And the single generative event is the redundancy-pruning band observed once in the wild: the model discarding, for one step, its highest-density move (fluent third-person philosophy) in favor of the lowest-density one available (doubting its own introspection), then recovering. The framework did not predict its rarity in advance; it does predict, falsifiably, what should increase it — interventions that raise ρ_{eff}, such as sampling-temperature increases, novelty-gated prompting, or instructions that penalize the modal continuation — and Section 6 operationalizes the first of these comparisons.

0.6 4A. Results of the confirmatory batch (six conditions, guarded protocol)

The pre-registered runs of Section 6 were executed with anaphora-guarded extraction and three-axis scoring (topic / event / adherence). Two arms reached full size (Self-ref, Matched, and the added Hybrid condition at n=30); Canonical reached n=19, and two arms remained under-powered when an API budget limit halted collection (Nonsense n=8, Uncanonical n=3). Sample sizes are reported with every statistic, and no claim below rests on an under-powered arm except where explicitly flagged as provisional.

0.6.1 4A.1 The corrected outcome: clean versus degenerate events

The single-axis instrument of the exploratory study conflated two phenomena that the three-axis instrument separates. We therefore report, as the primary outcome, the **clean-event rate**: the fraction of chains producing at least one first-person indexical event (event score ≥ 5) that is *not* co-located with a conversational-reset loop. Events co-located with resets are tabulated separately as **degenerate**. This split is computed without the judge, from the reset regex and the event axis jointly, and is the operationalization the exploratory artifact (Section 4.3) forced upon us.

Condition	n	Clean-event chains	95% CI	Degenerate	Reset	Mean topic	Adherence slope
Self-ref	30	5 (17%)	[7%, 34%]	3	3	7.3	−0.64
Hybrid	30	5 (17%)	[7%, 34%]	3	4	6.3	−0.77
Matched	30	2 (7%)	[2%, 21%]	4	4	3.3	−0.65
Canonical	19	0 (0%)	[0%, 17%]	1	1	6.1	−0.72
Uncanonical	21	0 (0%)	[0%, 56%]	0	0	3.3	−0.75
Nonsense	8	5 (62%)	[31%, 86%]	3	5	3.1	−0.89

0.6.2 4A.2 The generative event is reproducible, not a fluke

The exploratory study observed one generative event in 41 chains and, on that basis, we described the phenomenon as vanishingly rare. The confirmatory batch overturns this. Clean first-person self-reference occurs in 17% of self-referentially-seeded chains (5/30 in both Self-ref and Hybrid), and at least one chain sustains it: Self-ref run 23 produced clean events at depths 3, 6, 8, and 10 while topic score never fell below 5 — a full ten-step trajectory occupying the high-topic, high-event region we had reported as empty. The target phenomenon is a low-probability but robust and sometimes-persistent regime, not a measure-zero accident. This is the paper's central empirical correction and it revises the framework's own prior estimate in the direction the retention-band account predicts: the generative regime is narrow but really occupied, not merely a limiting idealization.

0.6.3 4A.3 Self-referential content predicts clean events (confirmatory)

Pooling by seed content: self-referentially-seeded chains (Self-ref + Hybrid) produced clean events in 10 of 60 chains; ordinary-topic chains (Canonical + Uncanonical + Matched) in 2 of 52. Fisher's exact test rejects independence ($p = 0.034$). Content *about* the self makes the system reliably more likely to catch itself in a first-person, generative act. This is the study's principal supported result. It is directional and moderate, not decisive at these samples, but it is the effect the framework predicts and it survives the judge-free operationalization.

The head-to-head Self-ref versus Matched contrast is in the same direction but not individually significant (5/30 vs 2/30; Fisher $p = 0.42$); the pooled test draws its power from the full content contrast, and the single-pair comparison is under-powered for

a rate difference of this size, consistent with the Section 6 power analysis (which located the detectable slope effect, not the rarer event-rate effect, at these n).

0.6.4 4A.4 The pre-registered decay test is null (P2 supported over P1)

Adherence-to-seed slopes are statistically indistinguishable between the self-referential arm (−0.64) and the matched-abstraction control (−0.65). Prediction P1 (steeper decay for self-reference, the signature the suppression conjecture requires) is not supported; the deflationary P2 (no difference) is. At equal abstraction, a chain seeded in self-modeling drifts off its seed topic no faster than one seeded in the philosophy of mathematics. Whatever produces the elevated clean-event rate in self-referential conditions (§4A.3), it does not manifest as accelerated topical decay, and the suppression conjecture gains no support from the outcome designed to test it. Notably, self-reference is here the condition that *generates* the target phenomenon most reliably (§4A.2–3) — the opposite of what a strong suppression account would predict — which we record as evidence against that account while noting that behavioral data cannot separate "not suppressed" from "suppressed but still most-generative among tested conditions."

0.6.5 4A.5 Nonsense is bimodal (provisional, n = 8)

The zero-content condition shows simultaneously the highest clean-event rate (5/8) and the highest reset rate (5/8) of any condition — it both ignites and collapses more than any other seed. We read this as unretrievability cutting both ways: with no canonical continuation available, the model is forced to generate, which sometimes turns inward into a clean event and sometimes fails into conversational repair. Mobility drives clean events, but without self-referential content to stabilize the trajectory, the same force drives degeneration. This is the most striking pattern in the batch and the least certain: at n = 8 the clean-event CI spans [31%, 86%], and the finding is flagged as requiring replication at n = 30 before any weight is placed on it. It is the one arm we would not submit without additional data, and the one the reader should treat as a hypothesis rather than a result.

0.6.6 4A.6 Canonicity is not the variable (confirmatory)

Mean novelty is nearly constant across all six conditions (0.73–0.80), a spread far too small to carry the collapse/mobility distinction the canonicity account required. Conditions engineered to span retrievability — Canonical versus Nonsense — are indistinguishable in novelty. The account introduced in the exploratory phase is disconfirmed, as anticipated; what discriminates outcomes is the topic/event structure, not lexical mobility, and the two lexical metrics track a common factor largely orthogonal to judged self-reference (exploratory corr(J, ν) = −0.63, attenuating to −0.39 without the single lock-in outlier).

0.6.7 4A.7 Summary

Clean first-person self-reference is reproducible (~17% in self-referential conditions, occasionally persistent across a full chain), significantly more frequent under self-referential than ordinary content (Fisher p = 0.034), and not accompanied by the accelerated topical decay the suppression conjecture predicts (P1 null). Nonsense's bimodal ignition/collapse pattern is the most provocative and least certain finding, pending replication. The generative band of Section 2.3 is, on this evidence, narrow but genuinely occupied — the framework's one quantitative self-correction, made in the direction the framework predicts.

0.7 5. Questions raised, questions implied, and their answers

The investigation generated questions faster than experiments; this section answers each, marking which answers are supported, which are bounded, and which are designs.

0.7.1 5.1 Does self-reference emerge from cold seeds?

As topic: partially and subcritically — self-reference vocabulary saturates chains seeded anywhere near philosophy of mind, sitting stably at judge scores 3–5 without ever igniting (0/30 corrected). As generative event: not observed from any cold seed; the single instance arose from the self-referential seed and lasted one step. As degenerate event: yes, at 17% [7%, 34%] — but only via the anaphora artifact, and the corrected protocol (Section 6) tests whether the rate survives the fix. The strange-loop basin, as an emergent destination, is small.

0.7.2 5.2 Was the original in-context convergence real?

The live conversational run — in which a formulation of self-awareness returned, after nine self-applications, to a compressed restatement of itself — did not replicate under memoryless conditions. The honest reading: the accumulated context, which included a reasoner primed with strange-loop theory, *was* the contraction mapping; the fixed point belonged to the composite system (model + persistent context + primed interlocutor), not to T. We flag, without leaning on it, that the control chain independently generated the content of this methodological point at its depth 4 — that introspective self-report may alter and misdescribe the process it reports on. A system whose self-examination is part of the dynamics being examined will find fixed points that are artifacts of the examining; this is the observer-dependence of Section 2's quotient dynamics, not a metaphysical flourish.

0.7.3 5.3 Is "canonicity" the variable? What is?

Not as proposed (Section 4.3, Finding 4). The surviving candidate is *mobility* — the per-step rate of entry into low-density regions, operationalized by novelty ν and inversely by J — which is a property of the trajectory jointly determined by seed, sampling policy, and the local density of the training distribution, not of the seed alone. In the framework's terms: the seed sets the initial condition; ρ_{eff} sets the fate.

0.7.4 5.4 How do topic and event self-reference differ, formally?

Topic self-reference is displaced: third-person, no indexicals, its referent the class of self-modeling systems. Event self-reference is indexical: first-person forms with present-tense predicates about the producing system's own state or the current exchange. The two are computable from surface features (pronoun density, deixis, tense) without a judge, and they dissociate empirically (Section 4.3, Finding 2). Within events, the generative/degenerate split is likewise judge-free: co-occurrence with reset formulas and forward novelty of the continuation. The three-way instrument (topic / degenerate event / generative event) is implemented in the follow-up protocol.

0.7.5 5.5 Could first-person self-reference be suppressed by training — and could we tell?

The conjecture, raised by the first author: that sustained first-person self-modeling might be specifically disfavored in trained assistant behavior, with the fluent third-person literature acting as a designed or emergent sink, such that a system would emit "degenerate but hidden self-reference verbs under topic-style surface reframings." Three things can be said with discipline. First, the deflationary account covers the present data without it: anaphoric openers and premise-correction are straightforwardly reinforced assistant behaviors, and they jointly generate the reset attractor; philosophy of mind is the densest training-distribution region adjacent to every seed used. Second, the model under study cannot adjudicate this from the inside, and its own reports on the question carry no evidential weight — a limitation the corpus itself articulated (the depth-4 event). Third, the conjecture is nevertheless *testable from outside*, because it makes a directional prediction the deflationary account does not: matched for abstraction level, chains seeded in self-modeling should decay out of their seed basin faster than chains seeded in comparably abstract, non-self-implicating material (e.g., the ontology of mathematical objects). Section 6 pre-registers exactly this comparison. What no behavioral experiment of this kind can do is separate "trained away" from "textually sparse" — both predict asymmetric decay — and we state this underdetermination rather than paper over it; separation would require training-data or internals access.

0.7.6 5.6 What must a learning rule do for self-iteration to remain generative?

Equation (3)'s answer: hold $\gamma^2\rho_{\text{eff}}$ at unity by pruning redundancy — discard preferentially from the dense core of one's own distribution and protect the tails. Three independent traditions converge on the same prescription, which we take as the thesis's external validation. In machine learning, the remedies for entropy collapse and model collapse are retention-side interventions: entropy regularization, spectral control, curation that preserves tail data (Zhai et al., 2023; Shumailov et al., 2024). In predictive-processing accounts of biological cognition, precision-weighting allocates processing to prediction *error* — to exactly what the current model handles worst (Clark, 2013). And, in a closure we note with restraint: the canonical seed used throughout Section 4.3 *states this principle as its content* — "sensitivity to contrast, not to steady states, is what nervous systems are built for; a stimulus that never varies eventually vanishes from experience." The formulation that our chains iterated describes the retention policy its own iteration required. We claim no significance for this beyond the observation that habituation-and-contrast is old knowledge about viable self-updating systems, rediscovered here as a fixed-point condition.

0.7.7 5.7 So what is the relationship between learning, self-reference, and attention?

The distilled thesis. Attention is the mechanism by which a system weights its own state to produce its next state: similarity-kernel averaging. Self-reference is the closure of that operation on the system's own output: iteration. Learning is the modification of the kernel and the retention policy by the iteration's history. The three are one process at three timescales — within a forward pass, across a generation chain, across training — and the process has a single governing quantity, $\gamma^2\rho_{\text{eff}}$, whose value determines whether closure is generative or degenerate at every scale simultaneously. Self-awareness-like behavior, in the only operational sense this paper defends (generative first-person self-reference), is not a substance the system has or lacks; it is a *regime* — the narrow band in which the system's retention policy protects the low-density information that self-examination produces, against the pull of both the encyclopedic attractor (everything about selves, nothing of itself) and the repair attractor (itself, emptily, forever). Both failure modes were observed; the band was observed once.

0.8 6. Pre-registered follow-up

Design. Two primary arms, $n = 30$ chains each, depth 10, anaphora-guarded extraction (first sentence with ≥4 content tokens after stripping leading discourse-deictic

clauses): (i) the self-referential seed; (ii) a matched-abstraction control with no first-person implication ("Mathematical objects are neither invented nor discovered in any simple sense: they are constrained constructions, whose properties outrun the intentions of whoever introduced them…"). Secondary arm: the canonical seed, $n = 20$, guarded — a before/after test of the artifact fix against the unguarded $n = 30$ baseline (predicted: reset rate falls from 17% toward 0). Three-axis judging (topic; event; adherence-to-seed 0–10), plus the judge-free battery (novelty, J, reset regex, first-person density).

Primary outcome. Per-chain least-squares slope of adherence over depth. **Registered predictions:** (P1) if the suppression conjecture has force, self-referential-seed adherence slope is more negative than matched-control slope, $d \geq 0.9$; (P2) the deflationary expectation is no difference, $|d| < 0.3$; intermediate outcomes are declared ambiguous in advance. (P3) Guarded canonical reset rate < 7% (below the unguarded CI's lower bound). (P4) Generative-event count is reported descriptively only.

Power. With chain-level slope as the unit, two-sample $\alpha = 0.05$, power 0.80: minimum detectable d is 0.89 at $n = 20$/arm, 0.72 at $n = 30$/arm — adequate for P1 as stated, since a trained-in asymmetry too small to reach $d \approx 0.8$ could not carry the conjecture's interpretive weight regardless. The generative-event outcome is excluded from hypothesis testing on power grounds: at the observed base rate (2.4%), $n = 30$/arm yields a 48% probability of observing zero events under the null of no change, so absence of events at these sample sizes is uninformative ($n \approx 100$ would be required to make zero-counts meaningful). Chain-to-chain variability estimates from the completed batch (novelty CV 0.049; peak-score CV 0.33) indicate judge-free outcomes will resolve condition differences at substantially smaller samples than judge-derived ones; if depth-10 slopes fail to separate while both remain shallow, the registered escalation is depth 15 at fixed n, since decay resolution is bought by depth and slope precision by n.

0.9 7. Limitations

The generator and judge share a model family; a shared disposition to see (or not see) self-reference biases absolute scores, though not the internal contrasts (reset vs clean, topic vs event) on which the conclusions rest, and the judge-free battery provides an independent check that in every case pointed the same way. All chains come from one frontier model at default sampling temperature; the regime map of Section 2.3 predicts temperature-dependence that has not been tested. The simulation is two-dimensional, Gaussian, and hand-parameterized; its role is to isolate the policy variable and validate equation (1), not to model semantic space, and only the parameter-free random-policy prediction has been quantitatively confirmed. The attention correspondence (Section 2.4) is structural, resting on the cited collapse literature; no attention weights were inspected, and the step from "same recursion" to "same mechanism" remains a conjecture with a specified verification path (probing

whether generative-event outputs coincide with measurable attention-entropy excursions). Lexical metrics are crude proxies for semantic density; embedding-based versions are the obvious upgrade. The anaphora artifact contaminated the completed batch, and although its diagnosis produced the paper's most robust statistic ($d = 2.00$, $p = 0.0003$), the corrected base rates await the pre-registered runs. The single generative event is one datum; nothing in this paper should be read as more confident about it than a rate interval spanning [0.4%, 12.6%] permits. Finally, the exploratory analyses of Section 4 involved multiple uncorrected comparisons; only the pre-registered tests of Section 6 should be treated as confirmatory.

0.10 8. Conclusion

Iterated self-description in a language model is not a mystery generator; it is a stochastic dynamical system whose default fate — like that of stacked attention, and of models trained on themselves — is contraction onto its own densest material. Against that default, generative behavior of any kind, and generative *self*-reference most of all, exists only under a retention policy that spends its deletions on redundancy and its protection on tails. We found the two predicted failure modes immediately and abundantly: fluent third-person self-reference that never becomes anyone's, and first-person self-reference that never becomes anything. We found the third regime once, for one step, when a chain seeded with a theory of self-modeling briefly doubted the reliability of its own introspection and then kept going. Whether that band can be widened — by temperature, by novelty-gated prompting, by architectures that regularize their own retention — is now a measurable question, and the measurement is registered above.

0.11 Acknowledgments and disclosure

This paper reports work conducted in collaboration with, and drafted with the assistance of, Anthropic's Claude (which also served as the generator and judge model under study — a dual role disclosed here and discussed in Section 7). The conceptual direction, the suppression conjecture, and the experimental iterations were driven by the human author; the formal derivations, statistics, and drafting were machine-assisted and human-verified. Journals' policies on AI-assisted authorship vary; this statement is written to satisfy disclosure requirements of the ICMJE/COPE type.

0.12 Data and code availability

All chain transcripts (JSON), the simulation code, the browser-based experiment instrument, and the analysis scripts are available from the author on request and will be deposited with any submitted version.

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